

**Daffodil International University**

Faculty of Science & Information Technology (FSIT)

Department of Computer Science and Engineering (CSE)

Course Code: CSE226, Course Title: Numerical Methods

1. Calculate the approximate value of, $I = \int_0^6 (e^x + 2x) dx$ by using **Trapezoidal**,

Simpson's 1/3, **Simpson's 3/8** and **Weddle's rule**. Also **Compute** the Exact Value and **Compare** the results & **Comment** on it.

2. **Estimate** the value of $y(6)$ for which the following data fits the Exponential Function, $y = a e^{bx}$

x	1	2	3	4
y	3	5	7	10

3. **Estimate** the value of $y(5)$ for which the following data fits the **Second Degree Polynomial** using the following data:

x	1	2	3	4
y	2	4	6	9

4. Use **Runge-Kutta 4th order** Method to solve the ODE, $10 \frac{dy}{dx} = x^2 + y^2$; $y(0)=1$ for the interval $0 \leq x \leq 2$.

5. **Solve** the system of Linear equations using **Gauss Seidel Method**:

$$x + 2y + 5z = 20$$

$$5x + 2y + z = 12$$

$$x + 4y + 2z = 15$$

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